

Jing-Zhang Open Pulse: A Verifiable AI Innovation Public Belt

> Core proposition: Jing-Zhang Open Pulse does not string AI devices into a showcase axis. It turns the heritage-park corridor into a civic loop: the three key areas respectively host trusted testing, open conversion and city experience; every component carries an ordinary equivalent service, a human owner, positive and negative evidence, a public receipt and a retirement decision. The proposal carries spatial judgment; GeoJSON, metrics, matrices and 40 gates carry recomputation and audit evidence.

One-page executive brief: civic loop and one-component/one-proof-mile

| Review question | Readable answer | Primary verification entry |

| --- | --- | --- |

| Brief alignment | Three scopes, three key areas, three zones/two wings, three positions, five functions and agent.1–agent.6 form one belt–three nodes–scenarios–operations chain; regional interfaces remain conceptual suggestions, not partnership commitments. |
compliancematrix.json, taskbook-crosswalk.json, regional-ecosystem.json, scope chapters |

| Originality | “Civic pulse + One Component, One Proof-Mile”: every scenario has an ordinary equivalent, an accountable owner, positive/negative evidence, a receipt and a retirement decision. | civic-pulse-protocol.json, proof-mile-delivery.json |

| AI and planning innovation | AI organizes questions, supplies explainable assistance and returns evidence; it does not replace planning, procurement, medical or legal decisions. Spatial moves land in three stations, slow routes, public space and bounded tests. |
scenario-operation-matrix.json, key-area-node-plans.json |

| Implementation feasibility | Move from ordinary service base to bounded prototype, supervised public window, then scale, repair or retire; eight action packages specify owners, dependencies, acceptance and stop rules. | operations-matrix.json,
construction-readiness.json, phasing chapter |

| Public interest | Residents, older/disabled people, caregivers, night workers, visitors, merchants, developers and maintainers retain ordinary routes; paper, human and account-free services cannot be replaced by AI. | persona-and-inclusion-matrix.json,
public-interest-audit.json |

| Risk and compliance | Provisional boundaries, unknown baselines, design targets and paper methods are not upgraded into official red lines, existing facts or approvals. Missing rights, fire, heritage, privacy, maintenance or takeover proof stops the concept. |
sources.json, risk.json, v2-evidence-gate-index.json |

| Expression completeness | Chinese and English proposals, five fixed review figures, A3/A0, offline HTML, recomputable metrics and three matrices share one spatial/version ledger; v3.3 subdivides buildings, green space, public rooms, scenario anchors, phases and
pending constraints without changing the prior area metrics. | manifest.json, metrics.json, geometry/.geojson, qa-readiness.json |

| Source-register completeness | The 59 unique source IDs in the Chinese proposal match the 59 records in package `sources.json` one-to-one; every record now provides a title, publisher or registering body, URL/path, access date, authority level, usable scope and
disabled scope. `data/sourceregistry.json` remains authoritative for formal/provisional use boundaries; package registration does not upgrade provisional material into formal evidence. | sources.json, data/sourceregistry.json, visual/assets/evidence-ledger.json |

v3.3 Three key areas: spatial moves before technology labels

| Area | Spatial problem | Design move and perceptible deliverable | Release boundary |

| --- | --- | --- | --- |

| Zhongzhi Garden AI Accelerator | Qinghe edge, low-carbon exchange, access and embodied-AI tests must not compete for the same public route. | Trusted test garden: connect green space, standards workshop, industry display, low-speed enclosed test court and
emergency-stop nodes into a low-disturbance loop back to the park. | Drainage, winter slip, fire, rights, data security and human takeover first; provisional polygons do not substitute for parcel conditions. |

| Beijing AI Origin Community | Near-campus transfer, talent life, open-source release and quiet daily use must coexist. | Open conversion street: stitch campus, park and neighbourhood movement with a release hall, IP advice, shade service bay and paper receipt point. |
Accessibility, night quiet, daily services and human windows come before small events or AI assistance. |

| Dazhongsi AI Industry Cluster | Station quadrants, enterprise exchange, commercial events and residents returning home can conflict. | Urban experience living room: station walking, four-quadrant crossings, quiet chain, short exhibitions and a capacity-reducible
international route form a recoverable public interface. | Fire, crowd, noise, drainage and rail/road coordination remain prerequisites; no event capacity or delivery schedule is promised. |

The three areas correspond to PROV-KEY-001, PROV-KEY-002 and PROV-KEY-003; these are provisional design constraints, not official red lines. The moves are conceptual suggestions for professional deepening after formal geometry, rights, controls, fire, heritage
and municipal conditions are available.

v3.3 Shortest implementation contract: an ordinary route and an exit at every step

1. Review: lock the official/provisional geometry version, field baseline, rights and accountable owner; if a critical input is missing, keep the state unknown.
2. Base: provide physical wayfinding, paper maps, human windows, seating/shade, water, drainage and account-free help first; ordinary service never waits for model recovery.
3. Test: operate only in supervised, time-bounded, low-speed windows with emergency stop and human takeover; record positive, negative, near-miss, complaint and group-difference evidence.
4. Decide: independent reviewers compare retain, repair, scale or retire; no scenario result is inherited by another component.
5. Receipt: publish version, sources, limits, deletion/withdrawal and maintenance records; on failure freeze scale and restore passive service.

Suggested first 90-day sequence (not a committed implementation schedule): days 0–30 only lock CR-01 evidence/site conditions and the field, rights and responsibility registers; days 31–60, after CR-01 passes, run the CR-02 passive-component mock-up and CR-03
wind/heat/water/lifeline coordination; days 61–90 consider one CR-04 bounded window only if CR-01–CR-03 pass, rights records are complete and staffed fallback exists. Any missing critical condition returns the work to review, repair or ordinary public service rather than

scale-up.

The machine-readable counts are [metric:civicpulsestagecount] stages, [metric:civicpulsestationcount] stations, [metric:proofmilecomponentcount] components, [metric:proofmilestationhandoffcount] station handoffs and [metric:riskclasscount] risk dimensions. These are package record counts, not field performance or approval conclusions.

The full formal dossier follows with design basis, three scopes, key areas, ecosystem and scenarios, land use, transport, municipal and blue-green systems, implementation, metrics, risk and references. Complete sources, standards, depth, data and stop rules remain in structured attachments.

Design Basis and Source List

This proposal follows the official open-call announcement, the repository site package and the agent taskbook. The machine-readable evidence chain is registered in `sources.json`, `standardmatrix.json`, `designdepthmatrix.json`, `compliancematrix.json`, and `visual/assets/evidence-ledger.json`. The most important source IDs are `[source:OFFICIAL-ANNOUNCEMENT]`, `[source:AGENT-TASKBOOK]`, `[source:SITE-PACKAGE]`, `[source:SOURCE-REGISTRY]`, `[source:PROCESSED-FACT-PACK]`, `[source:BOUNDARY-SOURCE]`, and `[source:KEY-AREA-SOURCE]`.

The package uses the repository's maintained provisional geometry where official boundary polygons are not available. `geometry/siteboundary.geojson` and `geometry/keyareas.geojson` therefore remain `officialboundary=false` and `provisionalconstraint`. They are suitable for design discussion, visualisation and self-checking, but not for a statutory red line, approval, precise land-area claim or government commitment. When official polygons arrive, all land use, roads, buildings, green space, public space, phasing and metrics must be recalculated.

Three-Level Scope Framework

The proposal works at three linked scales: the 43.6 km² coordinated research area, the 11.4 km² overall design area around the Jing-Zhang heritage park, and the 368.4 ha detailed-design area containing three key areas. These figures are taskbook scope descriptors, not a claim that the current provisional polygons reproduce official measurements. The three levels are cross-referenced in `compliancematrix.json` and `visual/assets/taskbook-crosswalk.json`.

The spatial idea is one belt, three nodes, many scenes, and a blue-green walking loop. “One belt” is a design method rather than a new legal boundary; the three nodes correspond to the three taskbook key areas; the loop joins slow movement, public space, water, shade and everyday services.

Scale Design question Evidence and decision
--- --- ---
Coordinated research How can knowledge, talent, compute, capital and public demand circulate? Eight-mechanism regional loop; conceptual only.
Overall design How do renewal, industry, transport, municipal systems and character fit together? Linked geometry layers, metrics and phasing.
Key areas How does each area become experienced, operated and reversible? Node concepts, 14 scenario rows and component library.

Coordinated Research Area: Industry and Future City Research

The research-area strategy is a public-facing innovation chain: university and community demand → open collaboration → enterprise conversion → ordinary public experience → evidence and international exchange. It is not an assertion that any named institution has agreed to participate. The regional mechanism diagram makes the proposed interfaces inspectable: Haidian universities and communities originate questions; Huairou contributes upstream science; Future Science City contributes engineering and energy testing; Beijing E-Town contributes manufacturing and deployment; the Jing-Zhang belt provides bounded public test windows; the broader Beijing-Tianjin-Hebei network compares and diffuses results.

The eight mechanisms are land, space, industry, capital, talent, compute, data and scenario. Each mechanism has an inbound need, an outbound return and an accountable future operator in `visual/assets/regional-ecosystem.json`. No partnership, funding, land allocation, or government commitment is asserted. The regional return loop is: originate → engineer → book → test → publish evidence → scale or retire.

The case comparison now uses six officially sourced public examples as mechanism references, not as local performance claims. The official pages are [source:CASE-HELSINKI-AI-REGISTER], [source:CASE-AMSTERDAM-ALGORITHM-REGISTER], [source:CASE-SINGAPORE-AI-VERIFY], [source:CASE-DECIDIM-BARCELONA], [source:CASE-UK-ATRS] and [source:CASE-SEOUL-AI-FOUNDATION].

Official example Verified mechanism Jing-Zhang interface Boundary Source
--- --- --- --- ---
Helsinki AI Register Public inventory and feedback route for city AI systems Status board with purpose, data boundary, accountable lead, state and complaint route Does not replace safety, accessibility, procurement or consent review [source:CASE-HELSINKI-AI-REGISTER]
Amsterdam Algorithm Register Plain-language public explanation of municipal algorithm use Preflight record linked to review, stop and redress paths Not Chinese legal compliance; protected details require review [source:CASE-AMSTERDAM-ALGORITHM-REGISTER]
Singapore AI Verify Standardised AI testing, open-source collaboration and assurance sandbox Booked human-led assurance window with model card, test note and rollback Not certification, procurement approval or permission to test residents [source:CASE-SINGAPORE-AI-VERIFY]
Decidim Barcelona Traceable digital and face-to-face participation infrastructure Contribution wall plus paper, walk-in and multilingual routes; publish what changed Digital participation cannot replace access for people without accounts or devices [source:CASE-DECIDIM-BARCELONA]
UK ATRS Standardised public record of why and how algorithmic tools are used Compact record for purpose, owner, data, human review, alternatives, limits and incidents Not a Beijing statutory requirement; local legal review remains necessary [source:CASE-UK-ATRS]

| Seoul AI Foundation | Institutional capacity linking research, public-service application, talent and global exchange | Named interface among research, talent, public service and exchange | No partnership, funding or city mandate is implied | [source:CASE-SEOUL-AI-FOUNDATION] |

The source IDs, access dates and use boundaries are recorded in `sources.json`. This is a mechanism comparison, not proof that Jing-Zhang has adopted or partnered with any of these organisations. The original six design patterns remain in `visual/assets/case-mechanism-matrix.json` as the local test layer.

The policy and enterprise-development interface is recorded separately in `visual/assets/case-policy-enterprise-crosswalk.json`: every case is linked to a policy tool, an enterprise question, a local scenario, acceptance evidence and a do-not-copy boundary. This moves the comparison from “seen a case” to “how a company enters a bounded testing, procurement, service and review loop” without turning a design suggestion into a招商, funding or government commitment. `visual/assets/policy-enterprise-playbook.json` turns that interface into 42 growth-stage cards, beginning with a public AI register and feedback card; `visual/assets/industry-validation-cases.json` adds three bounded validation windows; and `visual/assets/landmark-honor-crosswalk.json` links four public responsibility/contribution markers to the policy and enterprise interfaces. These are conceptual protocols, not procurement, funding or partnership commitments.

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

The overall design uses a renewal-first structure. Low-efficiency spaces are not assigned invented floor-area ratios or heights; instead, they receive a retain–renovate–demolish decision gate and a requirement for later official-control confirmation. In v3.3, the land-use layer remains an eight-class, non-overlapping design partition with recomputable shares, while the geometry package subdivides the unchanged building union into [metric:buildingunitcount] conceptual units, the green corridor into [metric:greensegmentcount] connected segments, and the public interface into [metric:publicspacepolygoncount] rooms plus [metric:publicscenarionodecount] scenario anchors. `metrics.json` exposes these counts and the unchanged area ratios; all polygons and points remain provisional and non-statutory. Roads, constraints and [data:geometry/phasing.geojson#PHASE-001] express the mobility network, pending professional inputs and conditional phasing.

Transport and municipal systems are designed as a continuous human network: rail-station interfaces, step-free crossings, low-speed logistics windows, cycle parking, emergency access, rainwater tree pits, shade, drainage maintenance and ordinary public use. AI may assist observation or scheduling, but a resident must still be able to walk, ask, complain, receive service and leave without an app, account or biometric identification.

Detailed Design of Key Areas

1. Zhongzhi Garden: autonomous-innovation accelerator

Zhongzhi Garden is a garden-like full-stack innovation district. Its public edge displays safety cases, standards work and low-carbon computing rather than turning the park into a private campus. A “stitch threshold” creates a step-free, audible and tactile crossing; a status board shows the current test window, accountable operator, stop condition and non-AI equivalent.

2. Beijing AI Origin Community

The Origin Community is a near-campus conversion and talent neighbourhood. The design stitches campus, park, enterprise and residential routes; creates an open publishing room; and reserves ordinary living, rest and care spaces. Results are presented as cleared models, diagrams and explainers, not as unlicensed historical or corporate imagery.

3. Dazhongsi AI industry cluster

Dazhongsi is an urban, internationally legible AI service and enterprise district. The station interface, four-quadrant pedestrian connections and retail/service edges are treated as one public sequence. Intelligent-device demonstrations are bounded by capacity, fire safety, resident access and a thirty-minute return-to-normal rule after an event.

Each key-area polygon is provisional and each node proposal is a concept for professional refinement. The node-level decision record is mapped to `[data:geometry/keyareas.geojson#PROV-KEY-001]`, `#PROV-KEY-002`, and `#PROV-KEY-003`.

AI Innovation Ecosystem, Personas, and AI+ Scenarios

The city is designed for more than one “AI user”. Personas include a university researcher, an SME founder, a community service worker, an older resident or carer, a visitor, a night-shift worker, an accessibility reviewer and a maintenance crew. Every persona has a non-digital fallback, a place to rest, a right to opt out and a visible complaint path.

The eight non-identifying public-user lenses are recorded in `visual/assets/persona-and-inclusion-matrix.json`: older/resident users; wheelchair, cane and guide-dog users; carers and children; night-shift workers; visitors; small businesses and community-service workers; developers/startups/university teams; and maintenance/operators. `userpersonacount` is a design-target count, not a population statistic or eligibility system. Each lens has a spatial response, offline fallback, test and safeguard, and the package prohibits facial recognition, individual trajectory retention, biometrics and commercial profiling. The matrix rows P-01 through P-08 remain conceptual until informed, opt-out and compensable participation is completed.

The package contains 14 scenario cards in `visual/assets/scenario-operation-matrix.json`. They cover walking-gap diagnosis, low-speed delivery robots, autonomous-mobility assistance, model red-teaming, open publishing, Jing-Zhang cultural wayfinding, health-service navigation, enterprise service, data-governance demonstrations, intelligent retail, public-safety review, an international AI week walk, rainwater/tree-pit maintenance, and a low-light quiet chain.

The matrix is a design-control instrument, not a promise of service levels. Every row states: spatial carrier, trigger, minimum data, RACI, response/recovery SLA, relative cost band, data retention, non-AI equivalent, stop condition, success metric and phase. The universal gates are: no digital precondition; human review cannot be removed; an accessible route remains continuous; a public status and complaint path exists; and stop/rollback has been tested.

To make a scenario card reproducible rather than merely readable, this package adds the independently designed `visual/assets/open-pulse-test-window-record.schema.json` and `visual/assets/example-s02-embodied-test-window.json`. The latter is a fully synthetic S02 low-speed-delivery sandbox test-window record: it puts the provisional road reference, no-app human alternative, minimum fields, visible stop and human spotter, public complaint route, maintenance responsibility and exit-asset plan in one record. Its result status remains `notrun`; the record is not robot performance or field acceptance. Any real pilot must create a new record and pass accessibility, safety, maintenance and public-review gates.

To address the structural-provenance and licence boundary exposed by #706, this version removes the peer-derived Relay Receipt files and replaces them with an independently designed Open Pulse bounded test-window record. The new record is organized around `window`, `placewindow`, `publicinterface`, `humancontrol`, `datacontract`, `observation`, `releasedecision` and `restoration`; it does not reuse the #426 schema skeleton. It remains a concept-only review interface, not deployment permission or field evidence.

Three industry validation windows: evidence before procurement or scale-up

`visual/assets/industry-validation-cases.json` defines three reversible, staffed validation windows for model safety, enterprise-service data governance and low-speed embodied AI. Each couples an enterprise question with a policy interface, acceptance evidence, a stop condition and a non-AI equivalent. The first window asks whether a supplier can reproduce safety, provenance, human review and rollback evidence; the second asks whether enterprise service can shorten a public process without unauthorised data or a binding AI decision; the third asks whether a robot can coexist with an accessible public route before scale-up. Two independent reviewers, a working manual route, clear rights, visible stop/takeover and a return to ordinary use are release gates. Unclear provenance, privacy spill, discriminatory routing, blocked accessibility, a severe near miss or overdue maintenance stops the window. These are conceptual test protocols, not local-company performance, procurement approval or deployment fact. [source:AGENT-TASKBOOK]

Cultural narrative, public markers, and long-term operations

The cultural narrative uses engineering proof, open collaboration and public reciprocity as a three-part experience. Four non-glitzy markers are proposed: the Century Engineering Questions Wall, the Open-Source Contribution Archive, the Urban Agent Responsibility Desk, and the Enterprise Safety-Governance Responsibility Desk. `visual/assets/landmark-honor-crosswalk.json` binds each to a spatial carrier, acceptance test, policy/enterprise interface and clearance boundary. They are conceptual nodes for heritage, public-art, management and rights review—not institutional endorsements, advertising sites or an approved event programme.

Node-level concepts and public-interest audit

`visual/assets/key-area-node-plans.json` makes the three key areas reviewable as sequences rather than slogans: arrival → status board → component → bounded operation → return to ordinary public use. Each node has components, an acceptance test and a professional gate for official parcels, ownership, roads, fire, ecology, heritage, drainage and municipal systems. It is a provisional concept for professional refinement, not an official parcel or construction drawing.

`visual/assets/public-interest-audit.json` deliberately leaves the baseline unknown until a survey is run. Older residents, disabled people, carers, night workers, children and guardians, visitors, small businesses and maintenance crews are named as participants. The audit tests step-free continuity, service without an app, complaint response, group differences and night light/noise. Participation has paper, staffed and multilingual channels; conflict resolution pauses the scenario, shows the disputed evidence, offers a non-AI equivalent and publishes remedy or withdrawal.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

The proposal avoids false precision. Building scale is expressed as a design relationship—heritage line, public node, quiet edge, enterprise interface and service courtyard—until official height, FAR, heritage and municipal constraints are supplied. Each future project must record whether an existing building is retained, renovated or removed, the reason, embodied-carbon implication, displacement risk, temporary public route and maintenance owner.

Transport, Rail, Municipal Infrastructure, and Public Services

The rail corridor is treated as a civic seam rather than a barrier. The design prioritises safe crossings, legible station approaches, cycle and wheelchair continuity, emergency access and low-speed delivery windows. Drainage is a visible operating system: rainwater tree pits, inlet inspections, post-storm work orders and no-device-dependent overflow paths are paired with any future sensing.

Public services are human-first. Health navigation is non-diagnostic; enterprise service uses authorised materials and human review; data demonstrations stay in a sandbox; and a network outage switches to printed guidance, staffed counters and ordinary public operations.

Urban Resilience, Embodied Intelligence, and Full-Life-Cycle Iteration

The future city is treated as a coupled human–environment–machine–asset system, not a list of smart objects. The climate-adaptation evidence boundary covers heatwaves, cold snaps, cloudbursts, drought, thunderstorms and freeze–thaw; air and microclimate cover ventilation, heat island, shade, PM2.5, dust and allergens; water and ecology cover terrain, pipes, infiltration, groundwater, water quality, biodiversity, bird collision and dark skies; human experience covers older people, children, disabled and pregnant people, night workers, noise, lighting, cognitive load, service access and belonging; embodied intelligence covers mixed traffic, emergency stop, tele-operation, charging/fire separation, information security, appeal and accountability; lifelines cover power, water, drainage, heat, communications and emergency response; construction covers reuse, materials, energy, carbon and waste; and operations cover asset ownership, inspections, desilting, spares, budgets, skills, work orders, renewal and exit. [source:RESILIENT-CITY-INFRASTRUCTURE-2024] [source:BEIJING-ACCESSIBILITY-REGULATION] [source:BEIJING-BIRD-BIODIVERSITY-2024] [assumption:A-BIODIVERSITY-LIGHT-001]

Low-regret comparison across all states

Five explicit design prototypes were compared: S0 minimal change, S1 technology-intensive, S2 ecology-intensive, S3 event-capacity-intensive and S4 balanced adaptation. The model uses 11 dimensions—thermal comfort, stormwater, air, human experience, robots, privacy, biodiversity, maintenance, carbon, flexibility and cost efficiency—five stakeholder weight profiles and eight stress states. A fixed seed `147228` produced 50,000 Monte Carlo draws with uncertainty in both weights and indicator responses. These are normalized decision scores, not field performance, climate forecasts or approval criteria. [assumption:A-RESILIENCE-MCDA-001] [metric:resiliencev13candidatecount] [metric:resiliencev13montecarlodraws]

| Prototype | Main advantage | Main cost or failure mode | Mean robust score | P05 | Win rate |

| --- | --- | --- | ---: | ---: | ---: |

| S0 minimal change | Low capital intensity and high reuse | Weak stormwater, maintenance, mixed-mobility and heat performance | 42.559 | 40.365 | 0% |

| S1 technology-intensive | Strong robot testing and digital base | Privacy, ecology, operational complexity and offline degradation | 54.090 | 50.613 | 0% |

| S2 ecology-intensive | Best stormwater, biodiversity, heat and carbon response | Higher event-capacity, separation and everyday maintenance pressure | 70.242 | 66.812 | 21.138% |

| S3 event-capacity-intensive | High event and logistics throughput | Hardscape, noise, heat, drainage and habitat costs | 46.579 | 42.994 | 0% |

| S4 balanced adaptation | Human experience, robot safety, maintenance, privacy and flexibility pass together | Higher capital intensity; S2 wins under a pure climate weighting | 72.518 | 70.252 | 78.862% |

S4 has mean regret of 0.314 and the lowest design score across eight stresses is 67.194. It passes five pre-set screening gates: core minimum ≥60, stress minimum ≥55, capital intensity ≤85, privacy ≥80 and human experience ≥75. S2 still wins for the climate-priority profile; that counterexample matters because S4 is a cross-group robust compromise, not the strongest option on every single dimension. [metric:resiliencev13selectedmeanscore] [metric:resiliencev13selectedp05score] [metric:resiliencev13selectedwinrate] [metric:resiliencev13selectedmeanregret] [metric:resiliencev13selectedminstressscore] [metric:resiliencev13hardgatespassed]

The eight stresses are a +2°C heatwave, +20% cloudburst, 30% irrigation reduction, 24-hour network outage, a threefold increase in robots, sudden event crowding, a 20% maintenance-capability reduction and winter freeze–thaw. Formal refinement must replace normalized inputs with seasonal CFD, measured wind/temperature/humidity/mean-radiant temperature, terrain and pipe models, infiltration, water quality, acoustic and light surveys, ecology baseline, energy, asset cost and real user tests. [assumption:A-CLIMATE-STRESS-001] [assumption:A-AIR-WIND-001] [assumption:A-DRAINAGE-SYSTEM-001]

S4 spatial operating system

1. Wind, heat and air: keep seasonal ventilation connections along the belt and three branches; do not form a fixed wind wall with continuous buildings, screens or dense planting. Compare leaf-on/leaf-off, prevailing wind, calm-pollution and heatwave cases before fixing trees, buildings or event structures. The city-scale ventilation network is background principle only, not an official Jing-Zhang corridor. [source:BEIJING-VENTILATION-NETWORK-2035] [source:BEIJING-METEOROLOGICAL-OPEN-DATA]
2. Rain and drainage: use the five-part chain source reduction → local detention → pipe connection → exceedance conveyance → post-storm recovery. The Haidian public sponge-city range of 75–80% annual runoff control is a screening reference only; final values require sub-area, terrain, soil, pipe, outfall and basement evidence. Underground space is never treated as a storm refuge. [source:HAIDIAN-SPONGE-CITY-PLAN] [source:BEIJING-FLOOD-PLAN-2021-2025]
3. The body first: a continuous step-free spine, heat refuge chain and quiet night chain have priority over exhibitions, delivery and robot charging. Older people, children, wheelchair/cane/guide-dog users, prams, cyclists and night workers must join route tests; a model score cannot overrule lived discomfort or a complaint. [source:BEIJING-ACCESSIBILITY-REGULATION] [source:WHO-HEAT-HEALTH-2026] [assumption:A-HUMAN-EXPERIENCE-001]
4. Embodied-intelligence levels: progress from world-model simulation to closed test site, low-speed scheduled geofence and only then limited public test. Identity, emergency stop, human takeover, appeal and withdrawal remain visible. Physical safety and GB/T 45502-2025 information security are both required; residents and disabled people are not free robot-training data. [source:BEIJING-EMBODIED-INTELLIGENCE-2025-2027] [source:SERVICE-ROBOT-INFOSEC-GB45502] [source:ISO-TR-4448-PUBLIC-MOBILE-ROBOTS] [assumption:A-EMBODIED-AI-SAFETY-001]
5. Ecology and night: use native, layered and drought-tolerant habitat, linked by ecological stepping stones. Review bird collision at glass interfaces; sensitive segments use dark-sky, low-glare, time-controlled light. No species increase or net ecological gain is claimed before a baseline survey. [source:BEIJING-BIRD-BIODIVERSITY-2024] [source:BEIJING-LIGHTING-GUIDE-2025] [assumption:A-BIODIVERSITY-LIGHT-001]
6. Maintenance is design: every tree pit, inlet, permeable surface, seat, light, sign, sensor and robot bay receives an asset ID, owner, inspection trigger, spare, service window, manual fallback and exit route before construction. Two consecutive overdue work orders or unavailable spares trigger removal of complex equipment in favour of passive, standardised, replaceable parts. [source:ASSET-MANAGEMENT-GBT33172] [source:RESILIENT-CITY-INFRASTRUCTURE-2024] [assumption:A-LIFECYCLE-MAINTENANCE-001]

Wind, water and health: a cultural boundary with an engineering gate

Cang feng ju qi (藏风聚气) and feng shui may appear as cultural landscape narratives, never as medical causation, air-quality evidence, hydrological law, engineering model or approval criterion. The current local baselines remain `unknown` and must be split into auditable gates:

- pedestrian-level wind-comfort acceptable-area ratio, tested for seasonal directions, calm conditions and leaf-on/leaf-off with calibrated CFD and field wind measurements; [metric:pedestrianwindcomfortacceptablearearatio] [source:LIU-URBAN-VENTILATION-2017]
- pollutant-stagnation hotspot count, requiring an emissions inventory, PM2.5 observations, boundary conditions and a professionally accepted air-age or ventilation-effectiveness method; [metric:pollutantstagnationhotspotcount] [source:MENG-WIND-HEAT-PM25-2022] [source:NOSEK-STREET-CANYON-2025]
- mean radiant temperature and continuous shaded accessible-route ratio, requiring seasonal solar/tree-canopy modelling, MRT measurement and accompanied-route tests; [metric:meanradianttemperaturebaselinec] [metric:continousshadedaccessiblrouteratio]
- blue-green accessible-route overlap, requiring field verification of entrances, gradients, continuity, safety, water quality and maintenance, without inferring health benefit from proximity; [metric:bluegreenaccessiblrouteoverlapratio] [source:WHO-URBAN-HEALTH-AND-GREEN]
- water-risk exceedance-route verification, requiring DEM, drainage, outfalls, soils, groundwater, water quality, design storms and calibrated surface-flow modelling before any flood-free statement. [metric:waterriskexceedancerouteverifiedratio] [assumption:A-DRAINAGE-SYSTEM-001]
- `designtarget` in `visual/assets/evidence-ledger.json` is a future review gate; the six `unknown` entries in `metrics.json` are the current evidence state. They may become `known` only when simulation input, calibration, field observation, formula, error and a responsible professional are all documented. `wind-health-validation-plan.json` and `wind-health-field-protocol.json` bind every observation to a geometry version, boundary case, source schedule, point ID, instrument, height, wind, PM2.5, thermal variables, canopy state and QC flag. Missing geometry alignment, calibration, safe field conditions, source/detection-limit information or pre-registered sampling stops interpretation and keeps the metric `unknown`. The identified Beijing climate-normal dataset, Haidian `1+21+65+100` network and Qinghe Station wind-monitoring route are lawful acquisition and context routes only, not site observations. [source:AIJ-CFD-PEDESTRIAN-WIND-2008] [source:AIJ-CFD-GUIDEBOOK] [source:ISO-7726-INSTRUMENTS-2025] [source:HAIDIAN-CLIMATE-NORMALS-DATASET-2025] [source:HAIDIAN-METEOROLOGICAL-NETWORK-2023] [source:QINGHE-STATION-WIND-MONITORING-2021]

Key-area stress tests and trigger-based rollback

- Zhongzhi Garden: treat the Qinghe edge as a shared wind, water, low-carbon-component and closed embodied-intelligence test; check drainage, winter slip, robot failure and data security before display.
- AI Origin Community: prioritise shade, quiet night learning, walk/cycle continuity and everyday service; robots operate only in windows that do not occupy the accessible spine.
- Dazhongs: resolve the station's four quadrants, cloudburst ponding, event crowds, ride-hailing/delivery, night noise and post-event recovery; event capacity yields to residents returning home, fire safety and exceedance conveyance.

Triggers convert future change into an executable fallback: heat warning reduces event capacity and opens refuge points; heavy-rain warning clears low points, checks inlets and opens exceedance routes; network loss stops autonomous robot movement while public facilities stay in manual mode; one non-takeover event, one blocked accessible spine or one serious near miss pauses robot testing; an asset without an owner or two overdue maintenance cycles blocks expansion. These are design triggers, not forecasts or statutory thresholds.

All-State Urban Operating System and Director's 30-Second Film

The all-state matrix tests the same public route under clear weather, rain, heat, night, network loss, power loss, sensor drift and failed robot takeover. In every state, tactile and printed guidance, emergency stop, lighting, help and staffed operation retain a minimum service. Embodied intelligence passes four gates—edge-compute, embodied-AI governance, privacy minimisation and a model card—before a small pilot; autonomy rate never replaces public access. Water follows source infiltration → local detention → pipe discharge → exceedance conveyance. Trees, winter slip, dismantlable materials, microgrid degradation, maintenance debt, asset responsibility and annual failure drills sit in one lifecycle loop. The dashboard distinguishes `known`, `designtarget`, `unknown` and `blocked`; a blocked

accessible spine, unsafe rain low point, failed emergency stop, unclear privacy boundary or overdue maintenance automatically degrades to manual mode and stops expansion. [source:NIST-HUMAN-CENTERED-AI] [source:ISO-13482-SERVICE-ROBOT-SAFETY] [source:ISO-55001-2024] [source:WHO-URBAN-HEALTH-AND-GREEN]

The short film is a communication layer, not engineering proof. Its five shots are deliberately traceable:

- 0–6 s, morning wind: a low tracking shot follows tree shadow, a tactile rail mark and a resident's wheelchair route; sound is leaves, not a synthetic “smart-city” swell. Overlay: `known / geometry version / no-app route`.
- 6–12 s, rain arrives: macro water runs from permeable paving into a rain-tree pit; a maintenance tag and inlet inspection appear, while a low point remains visibly reserved for exceedance flow. Overlay: `designtarget / drainage gate / post-storm work order`.
- 12–18 s, station crossing: at Dazhongsi's four-quadrant interface, an older resident crosses first, a delivery worker slows, and the status board shows a booked window. Overlay: `human priority / step-free chain / public stop`.
- 18–24 s, outage: the network indicator drops; the robot brakes and a human spotter takes over. Printed wayfinding and a staffed counter continue. Overlay: `blocked AI layer / ordinary service continues`.
- 24–30 s, night review: low-glare light, a quiet seat, an open contribution wall and a maintenance work order close the loop; the final frame says `scale, redesign or retire`. Overlay: `receipt / rights / evidence / rollback`.

Each shot maps to `visual/assets/evidence-ledger.json#film-storyboard` and `#film-production`, the relevant geometry layer, a sound cue, copyright record, operating state and fallback. The film must not imply official construction, measured health benefit, robot certification or a guaranteed future. [data:visual/assets/evidence-ledger.json#film-storyboard]

Blue-Green Network, Public Space, and Urban Character

The blue-green loop is a climate adaptation and social infrastructure, not a decorative green ribbon. Shade, quiet seats, permeable edges, rainwater storage, low-glare lighting and accessible routes are designed together. A component can fail digitally without failing physically: the rainwater tree pit still drains, the tactile route still works, and the seat remains a seat.

The identity system is two parallel rails, three open nodes and two switch diamonds. It turns the railway memory into a contemporary public grammar without using un-cleared historical images. The palette and wayfinding rules, including high contrast, tactile and audible alternatives, are machine-readable in `visual/assets/identity-system.json`.

This iteration also makes the identity direction inspectable as a vector mark in `assets/identity/open-pulse-mark.svg`: two parallel rails, three open nodes, two switch diamonds and bilingual naming remain legible at different scales. It is a concept asset authored by Xu Bingnan / Codex, not a registered trademark, government emblem or finished VI system; trademark, font, accessibility and public-communications review remain required before use.

Renewal Projects, Implementation Policy, and Phasing

Implementation is a sequence of reversible windows rather than a one-off technology launch:

- Read and prepare: confirm official polygons, survey accessibility, drainage and heritage; publish the rights ledger and baseline.
- Pilot and learn: open low-risk public-space components and two or three bounded scenarios with a staffed operator.
- Connect: add the regional return loop, enterprise and research interfaces only after the stop gates pass.
- Scale or retire: publish evidence, compare equity and climate outcomes, and either expand, redesign or remove the intervention.

The `scenario-operation-matrix.json` uses relative `low`, `medium` and `high` cost bands only; they are not budgets or procurement estimates. RACI and SLA fields identify the future owner needed for a formal implementation plan. If a responsible operator, safety case, rights record, drainage check or non-AI equivalent is missing, the scenario cannot advance.

Metrics, Area Recalculation, and Compliance Matrix

The current quantitative decision experiment compares baseline, adaptive, regenerative and stress states. The adaptive state has a model score of 97.39 with a minimum dimension of 91.14; the regenerative state is 97.26, baseline 95.86, and stress 91.78. These are design decision scores from the package model, not official statistics, a field measurement or a competition ranking. The historical 2023 index is 97.58 in the same model frame.

The preferred indicators are district income per capita, district GDP per capita and district health institutions per 10,000 people, with green coverage, physicians, wastewater treatment and water intensity as guardrails. Every number must be read with its source, year, unit, scope and limitation in `visual/assets/evidence-ledger.json`, `visual/assets/model-backtest.json`, `visual/assets/indicator-selection.json` and `metrics.json`.

Once official polygons replace the provisional geometry, the package must rerun area, ratios, key-area counts, object counts, accessible-route checks, blue-green continuity, drainage interfaces, land-use coverage, constraint interfaces and phasing. A passing self-check is evidence of package integrity, not evidence that a provisional concept is legally approved.

Risk, Copyright, and Compliance

The package includes a path-level `visual/assets/copyright-ledger.json` and an expanded `report/copyrightstatement.md`. Text, geometry, diagrams, offline HTML and figures are authored or derived in this worktree by 许丙南 / Codex. Official public data retain their source attribution. No third-party images, maps or remote runtime assets are embedded; local system fonts are used only for rendering and are not redistributed. The identity mark is a concept direction, not a registered trademark.

The rights ledger records authorship, source inputs, license scope, hashes and review limits for each asset. It is not a legal opinion. Any future team member adding an image, font, map, logo, data extract or event content must add a source, permission and hash before publication. Historical interpretation is expressed with text and self-generated diagrams unless a rights-cleared source is explicitly registered.

Public-interest checks cover accessibility, older residents, carers, night workers, privacy, conflict resolution and ordinary non-AI access. The package rejects facial recognition and individual trajectory retention for these scenarios. The stop condition is part of the design, not an afterthought: a failed safety, rights, drainage, access or resident-impact gate triggers human operation or withdrawal.

v1.6 Official Statistical Time Series and Decision Experiment

The v1.6 time-series section is retained in the Chinese source document and its machine-readable assets. It records the official statistical source boundary, the 2023 baseline frame and the four scenario comparison. It does not turn modeled outcomes into official forecasts. Readers should use the time-series figure and ledger to inspect source years, transformations, missing values and guardrails.

v1.7 Top-Rank Repair: From Concept to Reviewable Deliverables

This iteration directly repairs review gaps found in high-scoring repository submissions and the repository's detailed validator feedback:

- `visual/assets/taskbook-crosswalk.json` gives every agent.1–agent.6 requirement one unique section, artifact and acceptance test.
- `visual/assets/regional-ecosystem.json` and its figure turn regional cooperation into an eight-mechanism loop with explicit conceptual status.
- `visual/assets/scenario-operation-matrix.json` turns 14 scenes into spatial, operational and rollback records with RACI, SLA, cost bands and non-AI equivalents.
- `visual/assets/component-library.json` gives eight physical, legible and maintainable public-space components.
- `visual/assets/identity-system.json` and the identity figure provide a real, inspectable identity and wayfinding grammar.
- `visual/assets/copyright-ledger.json` makes authorship, sources, licenses, hashes and future-clearance duties auditable.
- `visual/assets/case-mechanism-matrix.json`, `key-area-node-plans.json` and `public-interest-audit.json` close the case-comparison, node-level and participation/equity evidence gaps without inventing a baseline or partnership.
- `visual/assets/operations-matrix.json` turns the design into eight accountable action packages—from official-boundary replacement and field baselines to embodied-intelligence windows, drainage maintenance and annual scale/redesign/retire decisions.
- `visual/assets/resource-accounts.json` makes five dependencies auditable: spatial rights, climate/water/ecology, model/data rights, human experience/participation, and assets/spares/maintenance. These are resource requirements, not budgets or funding commitments.
- `visual/assets/construction-readiness.json` adds five construction-to-opening gates and four passive/connected component interfaces; it does not add a construction drawing, quantity, budget or approval claim.
- `visual/assets/participation-cycle.json` records the return-pass process, participant lenses, current known/unknown states and the evidence required before a public window; it does not represent resident consent or a completed consultation.
- `proposal.md`, this English translation, two offline report pages, the offline visual index and `changelog.md` keep the bilingual review surface aligned.

These are reviewable design deliverables, not claims that a project has been commissioned. Their value is that a reviewer can find the requirement, inspect the evidence, test the stop gate and identify what must be confirmed before implementation.

The local review-surface record is `visual/assets/qa-readiness.json`. It records semantic HTML tables, locally linked bilingual figures and A3/A0 outputs, the offline-only runtime boundary, and the known limits that provisional geometry and unknown field baselines still require professional and public review.

The implementation handoff is deliberately inspectable: `operations-matrix.json` gives each package an owner, accountable reviewer, dependency, acceptance test, stop condition, maintenance rule and relative cost band; `resource-accounts.json` states which evidence and capacity must exist before a package is released. Together with the scenario matrix and the Open Pulse test-window record, these files form a reviewable chain from a public question to a bounded window, a human handoff, an evidence record and a scale/redesign/retire decision.

Construction readiness and continuous participation

The package now separates “ready to discuss” from “ready to build or open”. `visual/assets/construction-readiness.json` applies five gates: evidence/site lock; a public-route and passive-component mock-up; wind, heat, water and lifeline coordination; a bounded, staffed and time-limited window; and a public retain/redesign/retire decision [metric:constructionreadinessgatecount]. Each gate names entry evidence, outputs, pass and stop conditions, an ordinary non-AI baseline and a maintenance response. It explicitly rejects construction drawings, statutory plans, procurement specifications, cost estimates and implementation approval.

`visual/assets/participation-cycle.json` makes the return pass inspectable: synchronize inputs, reread changed requirements, compare peers, validate the package, publish a bounded review draft, establish field and public baselines, run a staffed window, return negative and distributional evidence, then retain, redesign or retire. Seven participant lenses [metric:participationlenscount] have paper, staffed, multilingual or no-app routes. The current package records repository sync, local validation and peer comparison as known review activities, while field baseline, resident consultation and public pilot remain `unknown` or `notstarted`; no consent, microclimate result or health outcome is inferred.

v1.8 Reviewer-Visible Evidence Layer

The v1.8 revision places the highest-impact taskbook evidence directly in `proposal.md` so a reviewer can audit it without opening every JSON attachment. The following crosswalk is an exact reading of the taskbook, not a new statutory claim.

| Taskbook layer | Jing-Zhang answer | Reviewable proof |

| --- | --- | --- |

| Three positions | Century Jing-Zhang culture belt; urban AI life-experience belt; AI-integrated innovation belt | Two-rail identity, no-app service, research → public test → evidence loop |

| Five functions | Full-stack autonomy; world-class ecosystem; AI+ scene paradigm; intelligent active city; global AI-governance voice | Zhongzhi Garden test gate, Origin open room, 14 scenarios, Dazhongsi public sequence, red-team and responsibility desk |

| Three areas / two wings | AI Origin Community; Zhongzhi Garden; Dazhongsi; Zhongguancun technology-service wing; Xiaoyuehe scene-empowerment wing | Provisional key-area geometry, node plans, regional ecology, blue-green and operations records |

The regional chain explicitly includes the Beijing North Latitude Community / Haidian universities and institutes, Huairou Science City, Future Science City, Beijing E-Town and the Beijing-Tianjin-Hebei network. Each interface has an input, an output, a future owner and evidence required before activation. All are conceptual suggestions; no partnership, funding, land allocation or government commitment is asserted.

| Regional interface | Input | Jing-Zhang output | Responsibility or evidence to confirm |

| --- | --- | --- | --- |

| North Latitude Community / Haidian universities | Resident questions, talent, open-source topics | Public scenario briefs, participation feedback, reproducible experiments | Community consent, participation support, privacy boundary |

| Huairou Science City | Upstream science and facility capacity | Explainable public displays and validation questions | Facility operator consent and data clearance |

| Future Science City | Materials, energy and engineering tests | Low-carbon components, drainage and robot-interface prototypes | Engineering, ecology and municipal review |

| Beijing E-Town | Manufacturing, industrialisation and deployment | Public-validation evidence after safety gates | Enterprise/park cooperation, procurement and liability boundary |

| Beijing-Tianjin-Hebei network | Diverse urban contexts and comparative feedback | Diffusion, reuse or retirement records | Regional authorities and operators |

Node and operations table

| Node | Human sequence | Embodied-intelligence boundary | Gate and acceptance test |

| --- | --- | --- | --- |

| Zhongzhi Garden | arrival → status board → stitch threshold → rain tree pit → booked test → park return | low-speed, booked window; visible emergency stop; never blocks the accessible chain | river/flood/traffic/ownership/fire review; a resident crosses, reads and leaves without an account |

| AI Origin Community | campus edge → open room → rights desk → contribution wall → learning → quiet residential edge | contributions can be attributed or withdrawn; data stays in a sandbox | parcel/heritage/displacement review; an older resident can use the route and service offline |

| Dazhongsi | rail arrival → four-quadrant crossing → quiet seat → short exhibition → international route → daily retail | event capacity can fall to zero; public broadcast and withdrawal bay remain visible | station/road/fire/crowd review; the route remains open and the space returns to ordinary use |

The 14-row matrix gives every scenario a spatial carrier, trigger, minimum data, RACI, response/recovery SLA, relative cost band, retention, non-AI equivalent and stop condition. In review shorthand: the walking and accessibility group pauses on a broken route; robot testing exits on a serious conflict or failed stop; red-team testing rejects unclear provenance; public publishing removes uncleared material; health navigation never diagnoses; events reduce to zero if fire, quiet or resident-access gates fail; rain and quiet-chain pilots freeze on overdue drains or threshold breaches.

| Scenario group | Spatial carrier | Trigger and minimum data | Responsibility and recovery gate | Non-AI equivalent / exit |

| --- | --- | --- | --- | --- |

| Walking-gap diagnosis and accessibility assistance | Roads and public space | Manual counts, obstacle log, accessible route | Transport/accessibility lead; seasonal check and 7–14 day complaint response | Human guide and physical signs; stop if the primary chain breaks |

| Low-speed delivery robot | Road constraints and withdrawal bay | Geofence, emergency stop, spotter, event log | Operator responsible, safety lead accountable; 24-hour review | Human delivery/handcart; exit after severe conflict or failed stop |

| Model red-team and data sandbox | Controlled test room and status board | Cleared sample, model card, evaluation log | Evaluation team and lab operator; publish window summary | Paper review/expert panel; reject unclear provenance |

| Open publishing, enterprise service and data room | Origin Community/service node | Authorised materials, public process, audit log | Community/service lead; 7-day record and correction route | Staffed counter and notice board; fuse on withdrawal |

| Health navigation and cultural wayfinding | Community node and green space | Non-diagnostic flow, cleared history, tactile/high-contrast guide | Service/park lead; staffed window always available | Human explanation, paper and audio; remove on diagnostic risk or unclear source |

| Intelligent retail and global AI week | Dazhongsi commercial sequence | Aggregate crowd, fire, quiet zone and event content | Event control/culture lead; 30-minute return and 14-day review | Ordinary retail/park opening; reduce capacity or cancel on threshold breach |

| Rain/tree-pit and quiet night chain | Green space, drainage and public space | Pre/post-rain ponding, light/noise log, manual inspection | Asset lead; 24-hour pre-rain and 48-hour post-rain closure | Manual clearing and physical wayfinding; freeze on overdue inlet or threshold breach |

Public-interest and case-comparison evidence

The baseline is deliberately unknown until surveyed. Older residents, disabled people, carers, night workers, children and guardians, visitors, small businesses and maintenance crews are named participants. Step-free continuity, service without an app, complaint response, group differences and night light/noise are audited through paired walks, staffed tests, stratified listening and public status boards. A conflict pauses the scenario, shows the disputed evidence, offers a non-AI equivalent and publishes remedy or withdrawal.

| Public-interest audit | Current state | Pre-pilot evidence | Conflict response |

| --- | --- | --- | --- |

| Step-free continuity and night access | Unknown; no invented baseline | Paired day/night walks, light and sound records | Pause, provide a manual route and publish repair |

| Service without an app | Unknown; included in service inventory | Paper, staffed, voice and multilingual route tests | Keep staffed window; digital layer cannot be a precondition |

| Group impact difference | Unknown; stratified survey required | Older/disabled/carers/night-worker/child/visitor/small-business/maintenance participation | Publish disagreement; do not hide it with an average |

| Complaint and response | Unknown; public status board to be established | Confirm clock and accountable owner for each case | Respond in 7–14 days or withdraw if unresolved |

| Privacy and rights clearance | Minimum data only; no individual trajectory | Source, consent, retention, deletion and hash review | Fuse on withdrawal or unauthorised access |

The case file compares six mechanism patterns—station-as-public-room, bounded living lab, blue-green service street, developer commons, quiet night network and heritage-as-method—rather than inventing international project facts. Each pattern says what to borrow, what not to copy, the Jing-Zhang test interface and what must be sourced before a formal citation. This preserves international communication while keeping the rights and evidence boundary auditable.

Long-term operation is seasonal rather than promotional: spring gathers public questions, summer opens bounded scenes, autumn hosts a developer/maintenance review, and winter publishes evidence and decides scale, redesign or retirement. Each season records RACI, rights clearance, maintenance, complaint handling and exit. This is an operational concept for professional, community and communications teams to deepen, not an approved event calendar or funding commitment.

Acceptance rule

The proposal is not considered implemented because it is written. It is reviewable because each task can be traced to a section, a spatial or operational artifact, a source/assumption, an acceptance test and a stop/rollback path. Official polygons, planning controls, ownership, municipal, heritage, traffic, weather, drainage and measured-experience data must replace the provisional and unknown fields before professional implementation.

Cultural “Feng Shui” Narrative and the Wind–Water–Health Evidence Gate

The expressions cang feng ju qi (藏风聚气) and feng shui may be used only as cultural landscape narratives linked to spatial memory. They are not medical findings, public-health causation, air-quality evidence, hydrological rules, engineering models or approval criteria. The narrative is therefore translated into six separate, auditable indicators whose current baselines remain `unknown` :

- pedestrian wind-comfort acceptable-area ratio, to be tested across seasonal wind directions and leaf-on/leaf-off states with calibrated pedestrian-level CFD and field wind measurements; [metric:pedestrianwindcomfortacceptablearearatio] [source:LIU-URBAN-VENTILATION-2017]
- pollutant-stagnation hotspot count, requiring an emissions inventory, boundary conditions, PM2.5 observations and a professionally accepted air-age or ventilation-effectiveness method; [metric:pollutantstagnationhotspotcount] [source:MENG-WIND-HEAT-PM25-2022] [source:NOSEK-STREET-CANYON-2025]
- mean-radiant-temperature baseline and continuous shaded accessible-route ratio, requiring seasonal solar/tree-canopy modelling, MRT measurements and accompanied-route tests; [metric:meanradianttemperaturebaselinec] [metric:continuoousshadedaccessiblerouteratio]
- blue-green accessible-route overlap, requiring field verification of entrances, gradients, continuity, safety, water quality and maintenance, with no direct inference of physical or mental-health improvement from proximity; [metric:bluegreenaccessiblerouteoverlapratio] [source:WHO-URBAN-HEALTH-AND-GREEN]
- water-risk exceedance-route verification, requiring a DEM, drainage network, outfalls, soils, groundwater, water quality, design storms and calibrated surface-flow modelling before any flood-free claim. [metric:waterriskexceedancerouteverifiedratio] [assumption:A-DRAINAGE-SYSTEM-001]

The `designtarget` entries in `visual/assets/evidence-ledger.json` are future review gates; the six `unknown` entries in `metrics.json` are the present evidence state. None of the three cited research cases is represented as Jing-Zhang CFD, monitoring or health evidence, and no published case percentage is transferred to this site. [assumption:A-AIR-WIND-001]

To keep “measure it later” from becoming an empty promise, `visual/assets/wind-health-validation-plan.json` defines an evidence contract for all six indicators. It binds each gate to a geometry version, wind/heat boundary cases, emission sources where relevant, field sampling, calibration error, accountable professional review and stop conditions. It is a validation protocol—not Haidian field data, CFD output, a health outcome or an engineering/approval document; the three ventilation papers supply method boundaries only and no case value is transferred.

This round adds `visual/assets/wind-health-field-protocol.json` to make “measure it later” a pre-registrable work package. Each observation must carry a stable `pointid`, `geometryversion`, timestamp, instrument, height, wind speed/direction, PM2.5, thermal variables, canopy state and QC flag. Wind, pollution and thermal observations are kept separate, with collocation/calibration, background and source schedules, same-point model–field alignment, and error reporting. The professional team must register point counts, repeats and final comfort thresholds before seeing the result; a convenience walk cannot stand in for a representative sample. Missing geometry alignment or calibration, unsafe field conditions, missing source or detection-limit information, or any attempt to turn “cang feng ju qi/feng shui” into causal evidence stops interpretation and keeps the metric `unknown`. [source:AIJ-CFD-PEDESTRIAN-WIND-2008] [source:AIJ-CFD-GUIDEBOOK] [source:ISO-7726-INSTRUMENTS-2025]

The data routes are separately recorded as identified but not downloaded: the Beijing public-data platform lists a Haidian ground-climate-normals dataset; a Haidian government report describes a `1+21+65+100` meteorological monitoring network; and a station-authority report documents wind monitoring at Qinghe Station. These are lawful acquisition, coordination and nearby-context routes only. They do not replace three-key-area observations or CFD calibration, and all six local metrics remain `unknown` until a versioned dataset is obtained and its station-to-site relevance is reviewed. [source:HAIDIAN-CLIMATE-NORMS-DATASET-2025] [source:HAIDIAN-METEOROLOGICAL-NETWORK-2023] [source:QINGHE-STATION-WIND-MONITORING-2021]

References

See `sources.json`, `standardmatrix.json`, `designdepthmatrix.json`, `compliancematrix.json`, `data/sourceregistry.json`, and the repository taskbook for the complete source and standard registry. The package intentionally preserves the distinction between official public information, user/repository-provided material, provisional geometry, modeled design scores, conceptual partnerships and future implementation requirements.